

Lightweight segmentation across medicine and agriculture

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This report covers two completed MK-UNet training experiments and the data checks needed to interpret them. Every reported score comes from a saved checkpoint and a per-image evaluation record.

Architecture and motivation

MK-UNet combines depth-wise convolutions at multiple kernel sizes with a U-shaped encoder and decoder. The model uses channel, spatial, and gated skip attention to refine the representation [1]. Our imported standard model contains 315,566 parameters. The main engineering attraction is a compact segmentation backbone that can be trained without a large pretrained encoder.

Experiment design

Experiment 1 uses the authors' ClinicDB split and the reference configuration. Experiment 2 trains a new model from scratch on CWFID [2]. Both use seed 42 and 200 epochs; the checkpoint is selected by validation Dice before test evaluation. The task and dataset change in experiment 2, while the architecture stays fixed.

Experiment	Test n	Dice / IoU	Best epoch
ClinicDB	62	91.28% / 85.19%	200
CWFID	21	87.98% / 78.59%	128

The ClinicDB run achieved 91.28% Dice and 85.19% IoU. Relative to the published 93.48% Dice reference [1], the difference is -2.20 percentage points. This comparison is descriptive: the publication averages five runs, whereas this study uses one seed and corrects evaluation and resizing behavior in the current public code.

[Code repository](#) | [Reproduction instructions](#).

What the implementation audit changed

Data integrity precedes metric comparison

The ClinicDB download preserves the authors' split and records file identifiers and SHA-256 hashes. CWFID's published split places image 028 in both training and test. We keep it in test and remove it from training. Eight remaining training images become validation samples, yielding 31 training, 8 validation, and 21 test images. Every pair is checked for matching dimensions; duplicate image hashes cannot cross splits.

Visual inspection showed that CWFID's binary masks use black for vegetation, so the loader explicitly inverts them. A regression test checks the vegetation fraction and prevents soil pixels from being mistaken for the segmentation target.

Evaluation and training corrections

The upstream loader returns PIL dimensions as width then height, while the evaluator reads them in the opposite order. Our evaluator uses native mask dimensions and preserves original ground truth. The upstream scale loop also overwrites its input between passes; our scale loop always starts from the original batch. These fixes improve protocol clarity but mean the execution is not byte-for-byte identical to the public runner.

The paper and repository defaults disagree. The reference configuration follows the paper's stated settings: 352-pixel input, AdamW learning rate and weight decay of 0.0001, batch 16, gradient clipping at 0.5, no augmentation, and three training scales. The agricultural run uses batch 8. ClinicDB uses float16 autocast with gradient scaling to fit the GPU; CWFID uses FP32. This precision difference and current CUDA PyTorch are recorded deviations.

With a fixed sigmoid threshold of 0.5, test Dice is 91.28% on ClinicDB and 87.98% on CWFID. The primary scores instead use the upstream per-image min-max normalization followed by 0.5 thresholding. This extra result exposes sensitivity to output scaling.

Bottleneck

Small parameter count does not guarantee proportionally low latency. Multiple depth-wise branches, attention operations, resizing, and GPU launch overhead remain. Model size and this study's wall time are observed quantities; mobile-device throughput and energy consumption were not measured.

Findings, limitations, and next experiments

On CWFID, the independently trained model achieved 87.98% Dice and 78.59% IoU on 21 held-out images. The target is all vegetation, merging crops and weeds. These results demonstrate that the architecture can be trained on an agricultural task; they do not establish zero-shot medical-to-agricultural transfer or robustness across farms.

What these experiments establish

The code can acquire and validate both datasets, train the original model, select a checkpoint, evaluate native-resolution masks, export predictions, and regenerate the submission documents. Saved manifests, logs, and checkpoint hashes tie each score to an actual run. They also make protocol changes visible instead of folding them into an unexplained performance difference.

What remains uncertain

A single seed cannot recover training variability. ClinicDB frames may be correlated within acquisition sequences, and patient-level separation is not verified. CWFID is small and comes from one field; its image-level split cannot establish unseen-farm reliability. Bootstrap intervals describe image resampling within the fixed test set, not uncertainty across new hospitals or farms.

Next study

The proposal develops spatially selective, bounded test-time adaptation. A useful next evaluation would keep source-trained weights fixed as a control, compare entropy adaptation against reliability-gated updates, and test whether rollback prevents degradation under sensor shifts. Hyperparameters must be selected on validation shifts; target test masks must never enter the adaptation objective.

Evidence and references

Execution detail: [experimental execution summary](#). Run evidence: [ClinicDB metrics](#) and [CWFID metrics](#).

[1] Rahman and Marculescu. MK-UNet. ICCV Workshops, 2025. [Paper](#); [official code](#).

[2] Haug and Ostermann. A Crop/Weed Field Image Dataset. ECCV Workshops, 2015. [Dataset paper](#); [data and split](#).

The public repository contains the configurations, manifests, per-image results, implementation, and reproduction instructions used in this study.